VENDOR PERFORMANCE\n& INVENTORY ANALYSIS

Final Exploratory Data Analysis Report

Charts and findings reproduced only from the supplied EDA__f.ipynb notebook.

Notebook finding	Value / result
Initial vendor-brand summary rows	10,692
Filtered rows used for main EDA	8,564
Top 10 vendors' purchase contribution	65.69%
Total unsold inventory capital reported	Notebook output reports approx. $2.71M
Mean unit purchase price by order size	Small: $39.07 | Medium: $15.49 | Large: $10.78
Profit-margin confidence intervals	Top-sales group: 30.74%–31.61%; low-sales group: 40.48%–42.62%

Purpose: summarize the notebook's existing charts, analysis tables, and stated findings for vendor selection, purchasing, pricing, and inventory management.

1. Executive Summary

- The notebook creates a vendor–brand summary joining purchase transactions, sales transactions, freight costs, and product pricing.
- The main EDA filters to GrossProfit > 0, ProfitMargin > 0, and TotalQuantity > 0. The displayed filtered dataset contains 8,564 rows, compared with 10,692 rows in the initial summary.
- The top ten vendors account for 65.69% of purchase contribution in the notebook's displayed Pareto analysis.
- The notebook's bulk-purchasing boxplot shows a lower central unit-price distribution for larger orders, although there are substantial outliers.
- The notebook reports higher average profit margins for the low-sales quartile than the top-sales quartile and displays their confidence-interval comparison.

Priority recommendations based on notebook findings

Area	Action to consider	Notebook evidence
Supplier concentration	Review alternate sourcing and dependency exposure.	Top 10 purchase contribution = 65.69%.
Inventory	Investigate unsold capital and vendors with turnover below 1.	Notebook computes unsold inventory value and lists low-turnover vendors.
Pricing / promotion	Test selected low-sales, high-margin brands in controlled campaigns.	Notebook identifies target brands using 15th/85th percentile thresholds.
Bulk purchasing	Compare savings with storage, carrying cost, and demand risk.	Unit price is lower on average in the Large order group.
Data quality	Validate zero denominators, negative gross profit, outliers and extreme turnover.	Initial summary statistics show negative/zero and extreme values.

2. Data Preparation and EDA

The notebook explores the vendor_sales_summary table, containing fields such as VendorNumber, VendorName, Brand, Description, PurchasePrice, ActualPrice, Volume, TotalPurchaseQuantity, TotalPurchaseDollars, FreightCost, TotalQuantity, TotalSalesDollars, TotalSalesPrice, TotalExciseTax, GrossProfit, ProfitMargin, StockTurnover, and SalestoPurchaseRatio.

Step	Notebook operation	Interpretation
Summary statistics	df.describe().T	Shows ranges, means, quartiles and extremes for numeric fields.
Initial distribution plots	Cell 58	Histograms for numerical columns before filtering.
Initial outlier boxplots	Cell 59	Boxplots reveal extreme values across numerical fields.
Filtering	GrossProfit > 0; ProfitMargin > 0; TotalQuantity > 0	Removes loss-making, invalid-margin and zero-sales-quantity rows from later analyses.
Post-filter distributions	Cell 63	Histograms after the notebook's filters.
Categorical counts	Cell 64	Count plots for VendorName and Description.
Correlation	Cell 65	Heatmap for numeric columns in the filtered dataset.

Original EDA charts — distribution and quality checks

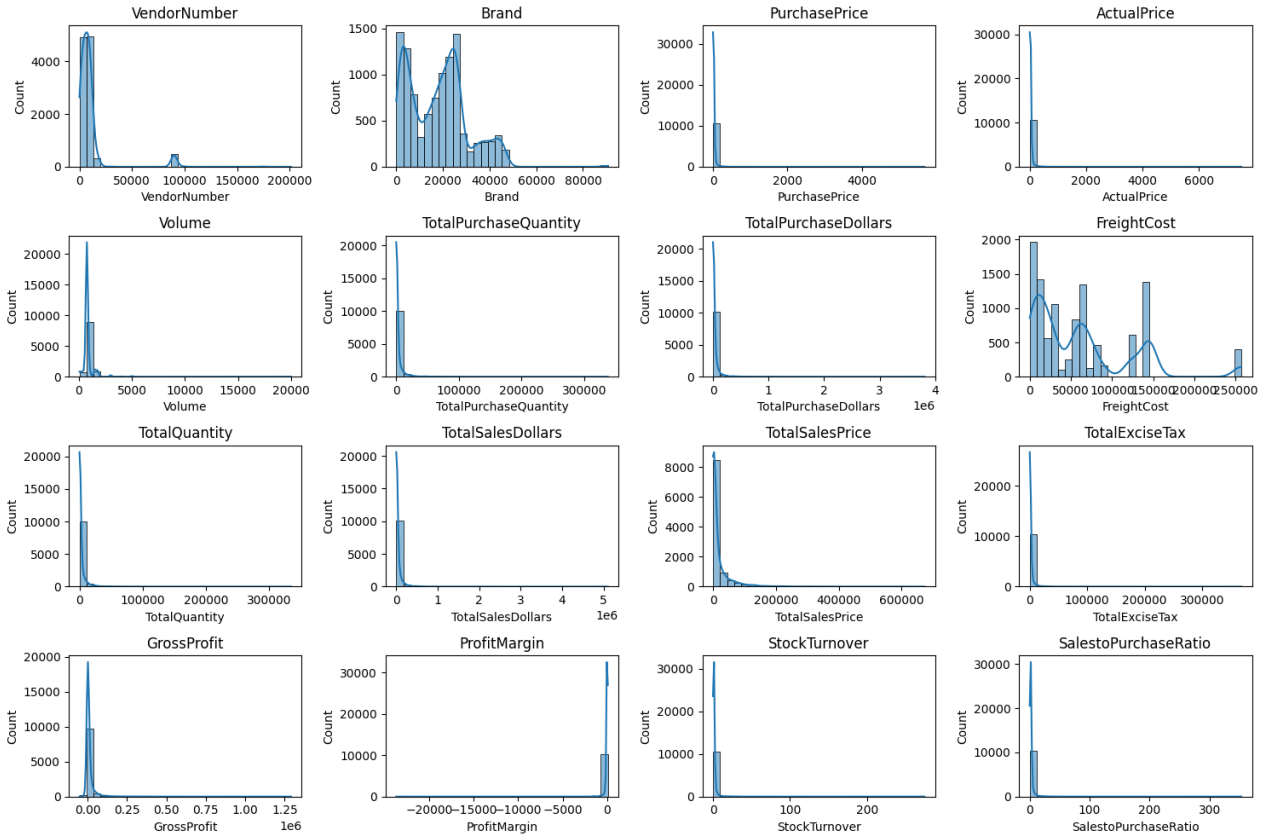


Figure 1. Original notebook chart: numerical distributions before filtering (cell 58).

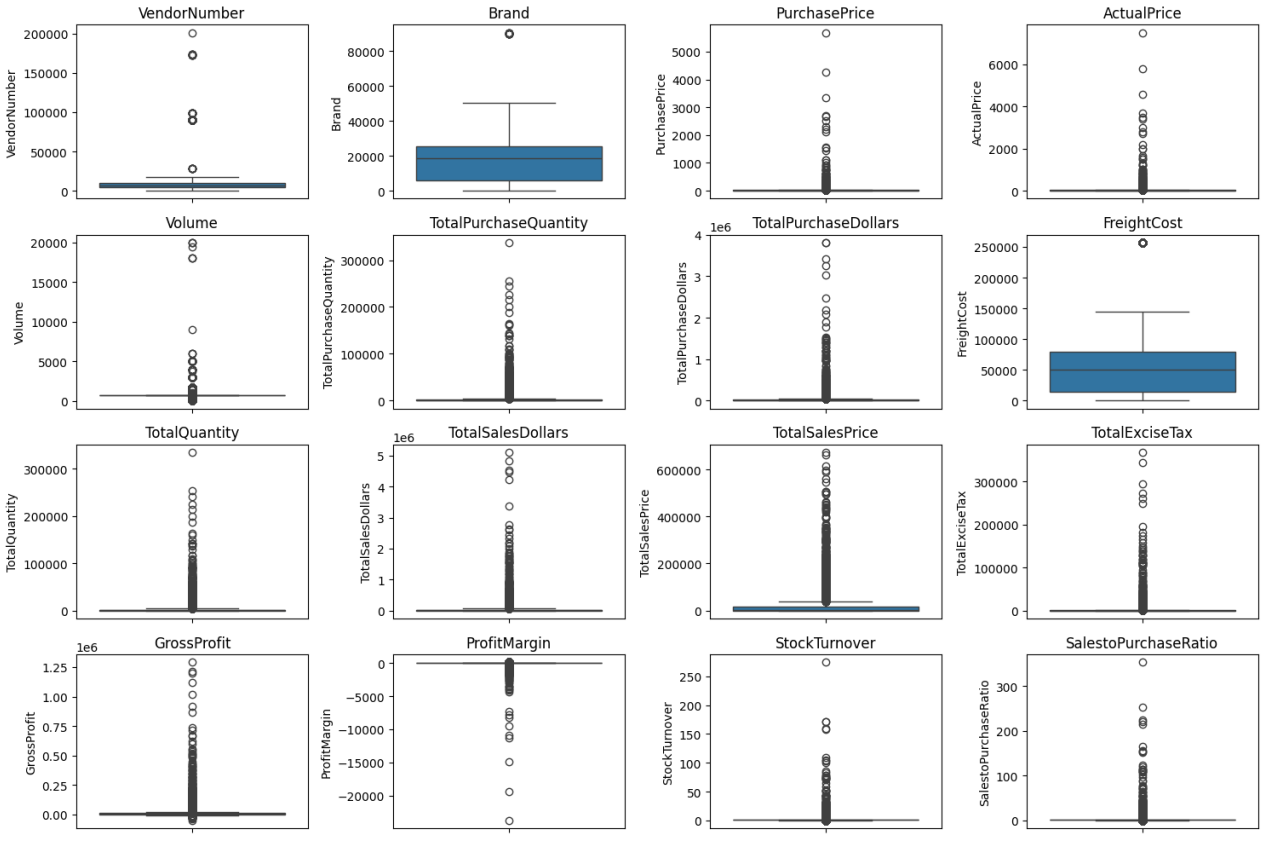


Figure 2. Original notebook chart: numerical boxplots before filtering (cell 59).

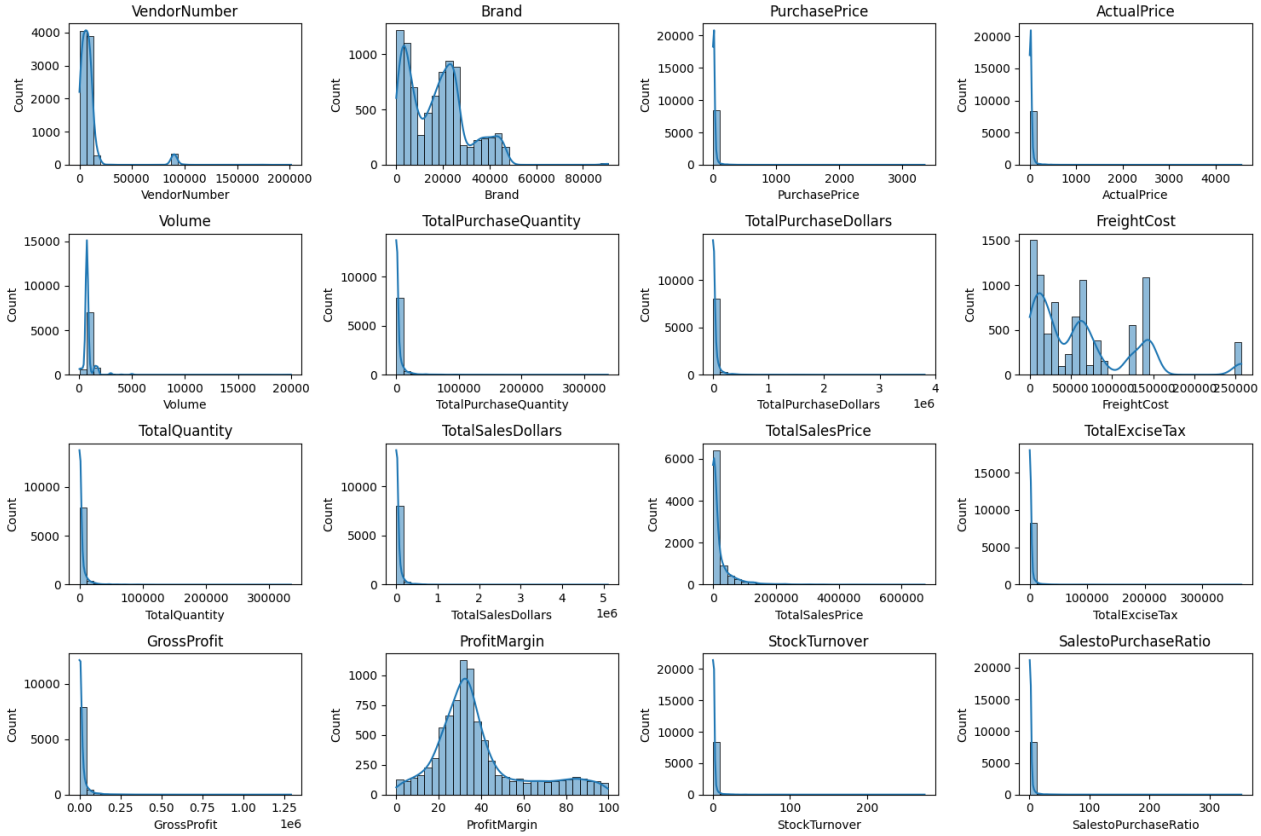


Figure 3. Original notebook chart: numerical distributions after filtering (cell 63).

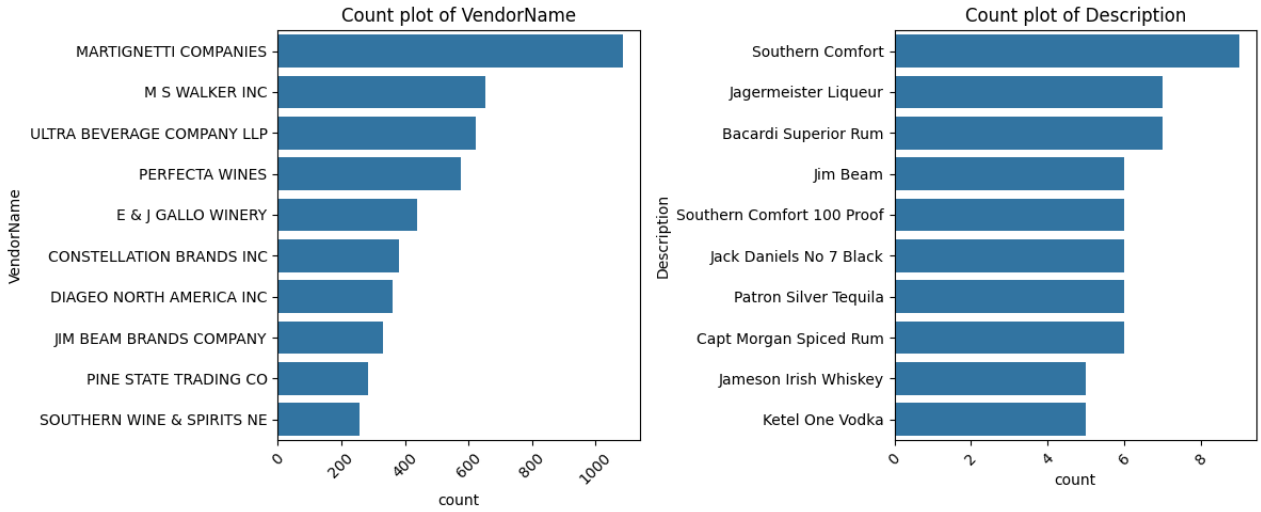


Figure 4. Original notebook chart: count plots for vendor and product description (cell 64).

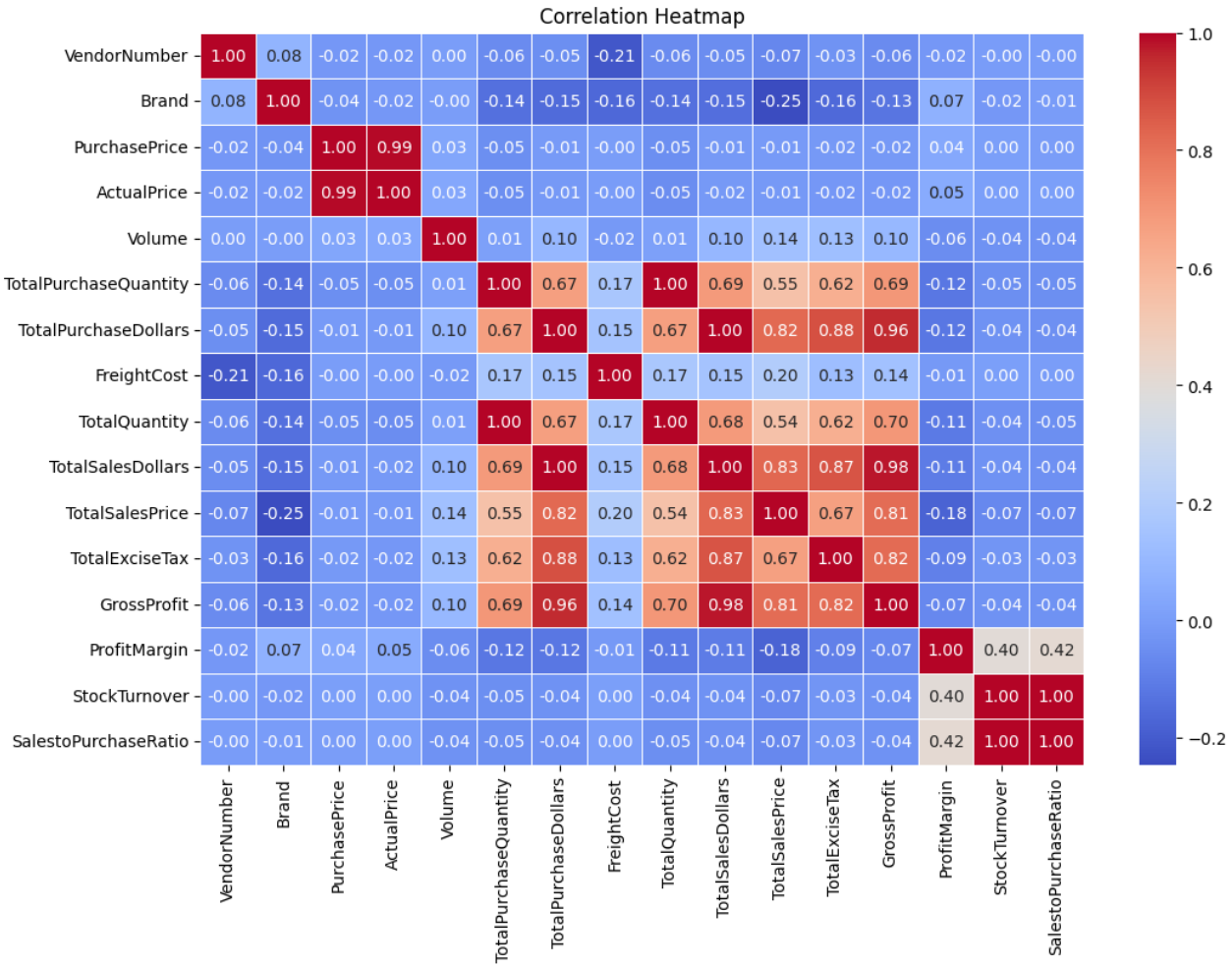


Figure 5. Original notebook chart: correlation heatmap (cell 65).

3. Brands for Promotional or Pricing Adjustments

The notebook aggregates brand-level TotalSalesDollars and mean ProfitMargin. It sets the low-sales threshold to the 15th percentile of brand sales and the high-margin threshold to the 85th percentile of mean brand margin, then highlights target brands meeting both conditions.

Notebook criterion	Threshold shown in notebook
Low sales threshold	$560.299
High profit-margin threshold	64.9702%
Target group	Brands at or below low-sales threshold and at or above high-margin threshold

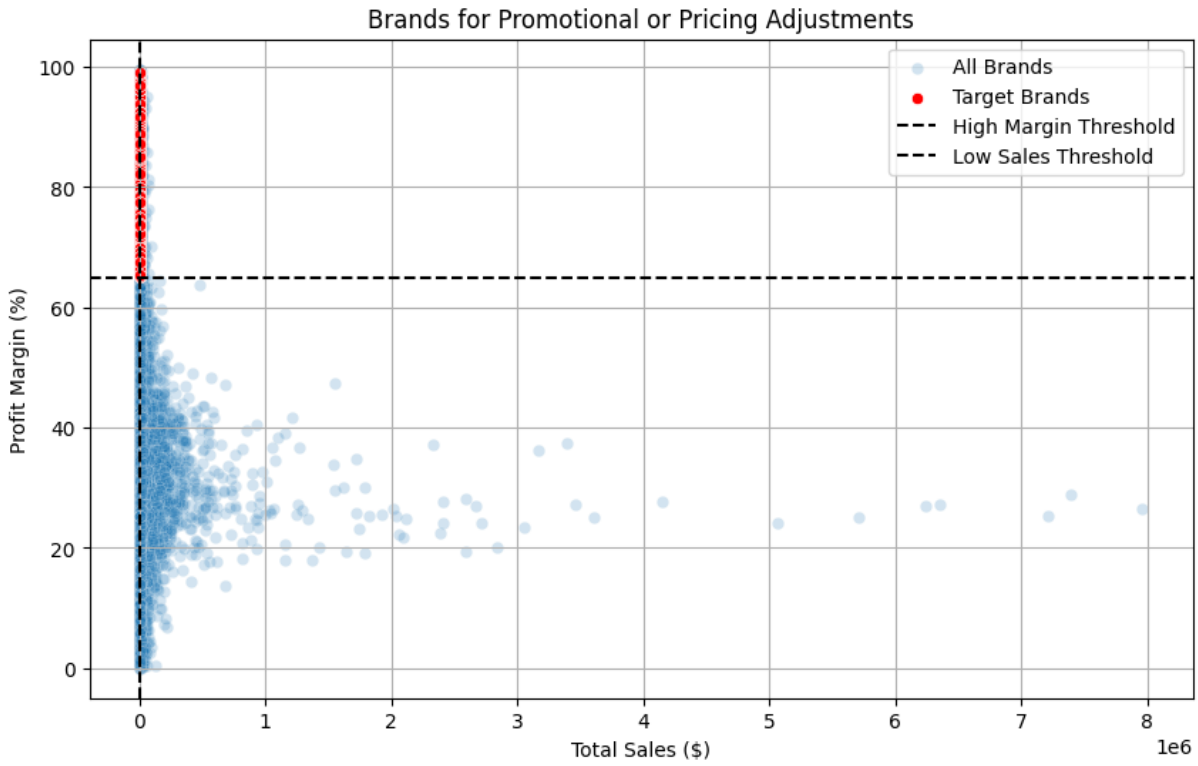


Figure 6. Original notebook target-brand scatterplot (cell 73).

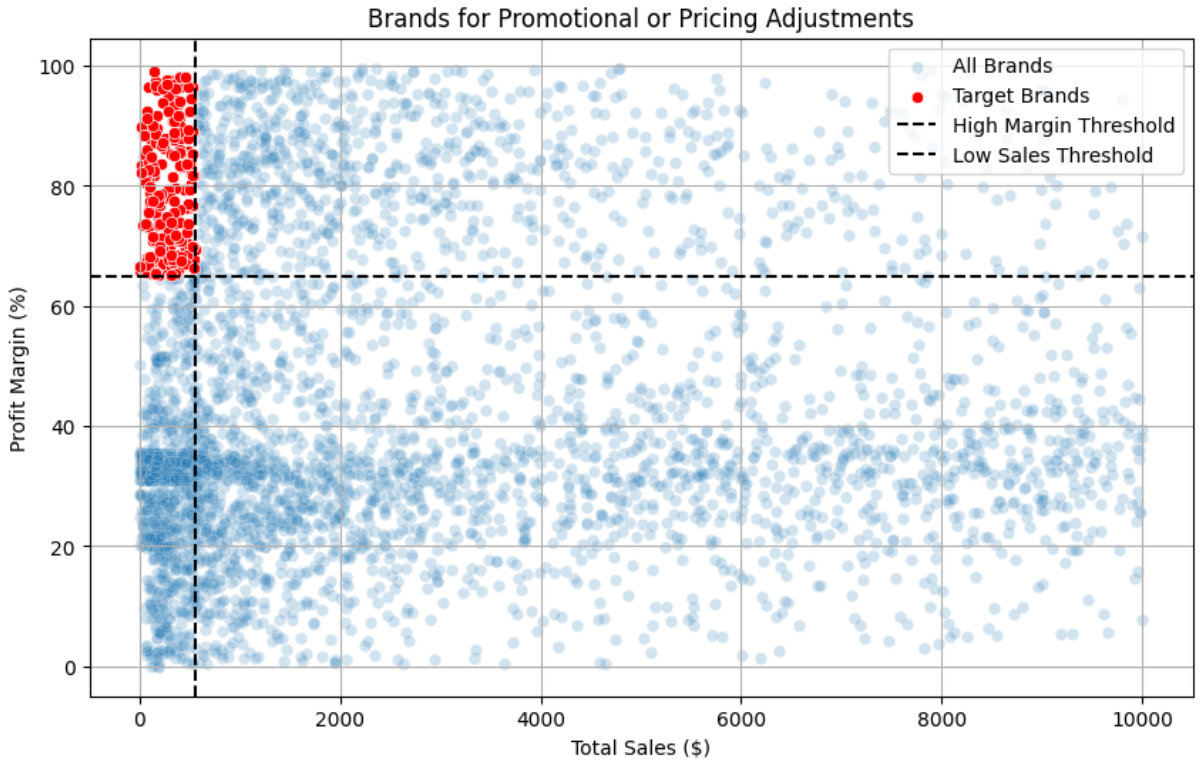


Figure 7. Original notebook scatterplot after limiting the all-brand display to sales below $10,000 (cell 75).

- Use the target brands as candidates for controlled promotion tests rather than assuming that a discount will increase profit.
- Track incremental sales, gross profit dollars, sell-through, and margin against a control group.

4. Vendor and Brand Sales Performance

The notebook ranks the top ten vendors and top ten brands by TotalSalesDollars.

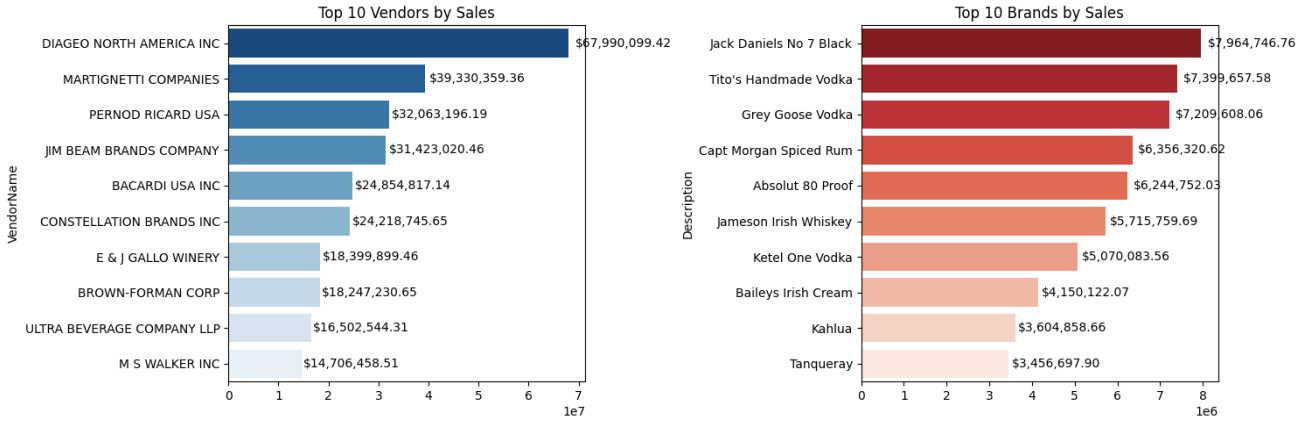


Figure 8. Original notebook chart: top 10 vendors and top 10 brands by sales (cell 79).

Top vendors shown in chart	Top brands shown in chart
DIAGEO NORTH AMERICA INC	Jack Daniels No 7 Black
MARTIGNETTI COMPANIES	Tito's Handmade Vodka
PERNOD RICARD USA	Grey Goose Vodka
JIM BEAM BRANDS COMPANY	Capt Morgan Spiced Rum
BACARDI USA INC	Absolut 80 Proof
CONSTELLATION BRANDS INC	Jameson Irish Whiskey
E & J GALLO WINERY	Ketel One Vodka

BROWN-FORMAN CORP	Baileys Irish Cream
ULTRA BEVERAGE COMPANY LLP	Kahlua
M S WALKER INC	Tanqueray

The figure is the notebook's own combined bar chart. Use the chart for ranking and relative scale; vendor sales alone should not be treated as a profitability score.

5. Purchase Contribution and Supplier Concentration

The notebook groups purchase dollars by vendor, calculates PurchaseContribution%, sorts the result, and plots the top ten vendors with cumulative contribution.

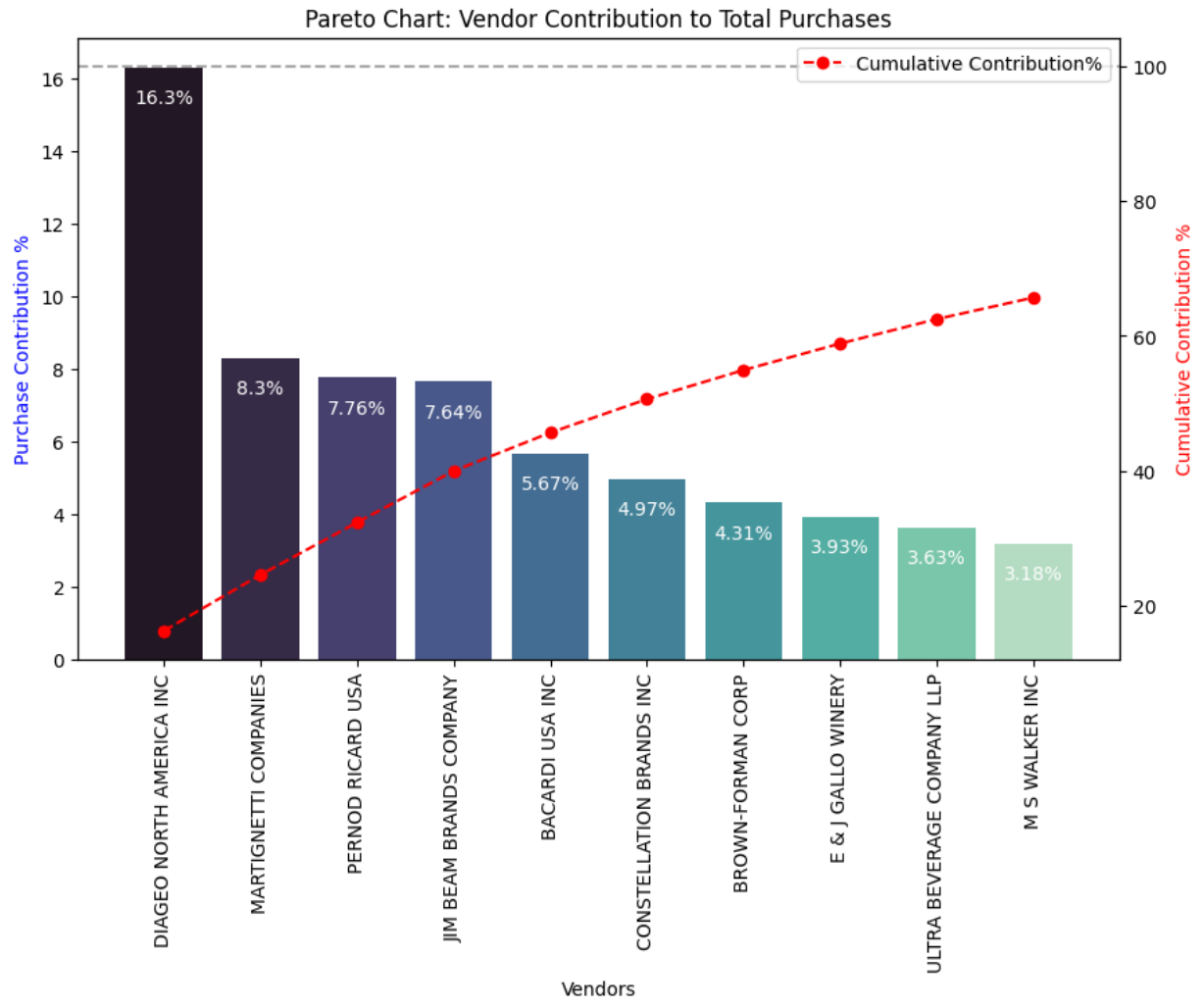


Figure 9. Original notebook Pareto chart: vendor contribution to total purchases (cell 90).

Vendor in chart	Purchase contribution shown (%)
DIAGEO NORTH AMERICA INC	16.30
MARTIGNETTI COMPANIES	8.30
PERNOD RICARD USA	7.76
JIM BEAM BRANDS COMPANY	7.64
BACARDI USA INC	5.67
CONSTELLATION BRANDS INC	4.97
BROWN-FORMAN CORP	4.31
E & J GALLO WINERY	3.93
ULTRA BEVERAGE COMPANY LLP	3.63
M S WALKER INC	3.18

The notebook reports that the top ten vendors contribute 65.69% of total purchase dollars. The chart's cumulative line makes the concentration visible. Supplier dependency should be evaluated alongside product availability, substitutability, lead time, and vendor service performance.

6. Bulk Purchasing and Unit Price

The notebook calculates UnitPurchasePrice as TotalPurchaseDollars / TotalPurchaseQuantity and divides records into three quantity-based quantile groups using pandas qcut: Small, Medium, and Large.

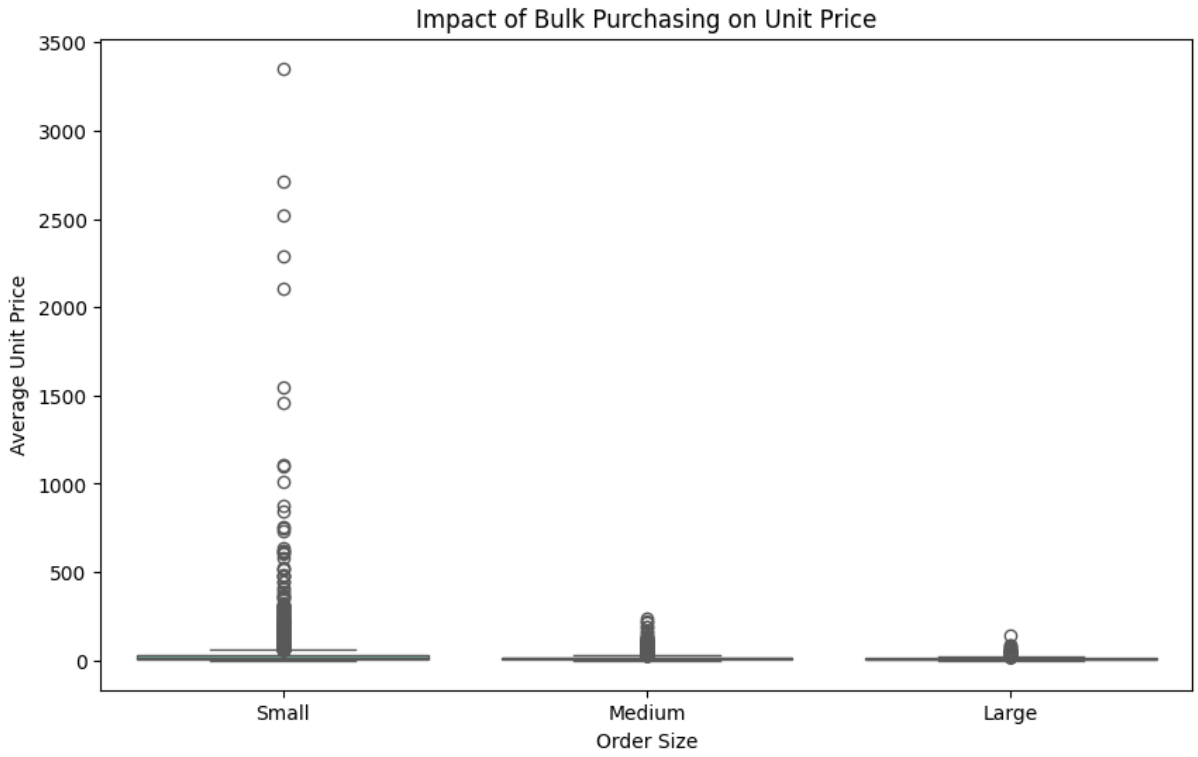


Figure 10. Original notebook boxplot: impact of bulk purchasing on unit price (cell 98).

Order-size group	Mean unit purchase price in notebook
Small	$39.07
Medium	$15.49
Large	$10.78

- The Large group has the lowest mean unit purchase price in the notebook output.
- The boxplot also shows extreme outliers, especially in the Small group; investigate product mix and premium products before generalizing.
- The notebook does not establish a single optimal purchase volume. Assess landed cost, carrying cost, storage, expiry, demand forecasts, and stockout risk before changing order sizes.

7. Slow-Moving Inventory and Unsold Capital

The notebook identifies vendors whose average StockTurnover is below 1 and calculates UnsoldInventoryValue as (TotalPurchaseQuantity − TotalQuantity) × PurchasePrice. It reports total unsold capital of approximately $2.71M.

Analysis	Notebook method / result
Low-turnover vendor list	Filter StockTurnover < 1, group by VendorName, sort ascending, show first 10.
Unsold inventory estimate	(TotalPurchaseQuantity − TotalQuantity) × PurchasePrice.

Total unsold capital	Approximately $2.71M as reported by the notebook output.

- Prioritize reconciliation against on-hand stock, returns, transfers, shrinkage, and the approved inventory valuation method before treating the estimate as a financial balance.
- Use SKU-level age, last-sale date, available quantity and carrying value to determine specific actions.

8. Profit-Margin Confidence Intervals and Statistical Test

The notebook separates records into the upper sales quartile (top-performing) and lower sales quartile (low-performing), then compares ProfitMargin distributions. The final chart below is the original notebook chart and is included without redrawing or replacing it.

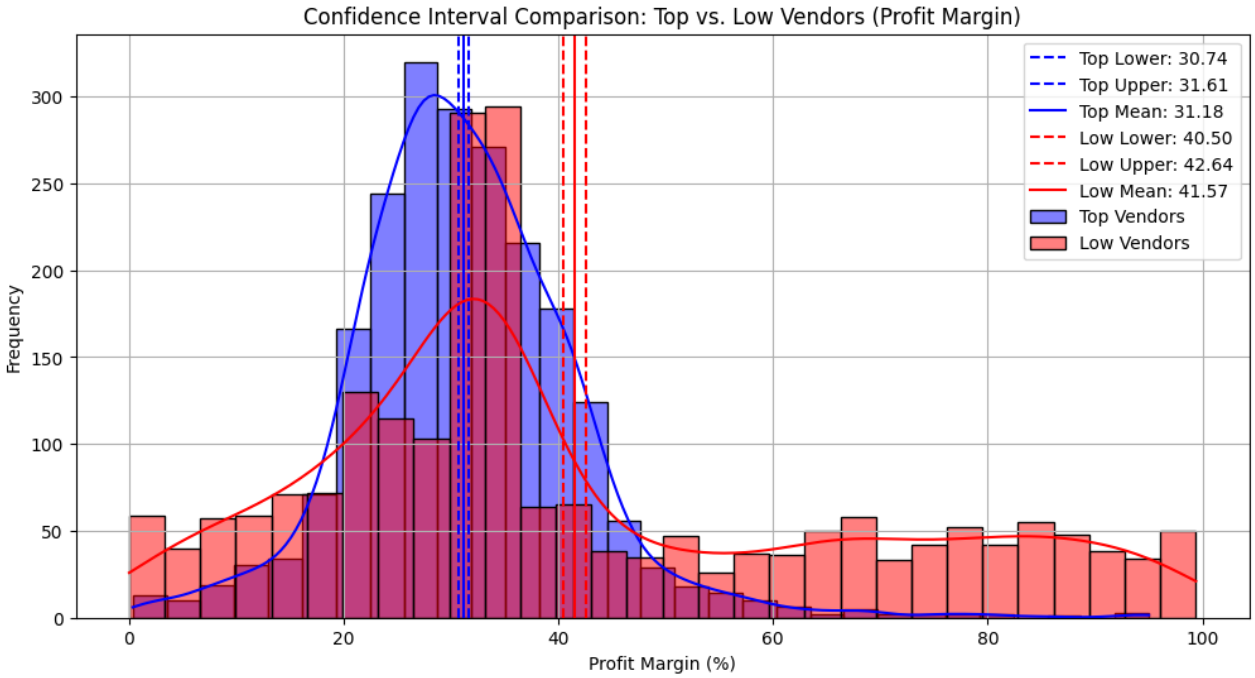


Figure 11. Original notebook confidence-interval comparison for top- vs. low-sales vendors (cell 110).

Group	95% confidence interval for mean profit margin
Top-sales quartile	30.74% to 31.61%
Low-sales quartile	40.48% to 42.62%

The notebook narrative states that the low-sales group's confidence interval is higher than the top-sales group's. The subsequent Welch two-sample t-test is documented in the notebook as testing whether the mean profit margins differ. Statistical significance does not explain the cause of the difference; investigate product mix, price positioning, cost, and volume.

9. Correlation Insights from the Notebook

Use Figure 5—the notebook's original correlation heatmap—as the source of correlation values. The notebook narrative specifically notes the following relationships:

Variable pair	Correlation stated in notebook	Interpretation in notebook
PurchasePrice vs TotalSalesDollars	-0.012	Weak negative relationship.
PurchasePrice vs GrossProfit	-0.016	Weak negative relationship.
TotalPurchaseQuantity vs TotalSalesQuantity	0.999	Very strong positive relationship.
ProfitMargin vs TotalSalesPrice	-0.179	Weak negative relationship.
StockTurnover vs GrossProfit	-0.038	Very weak negative relationship.
StockTurnover vs ProfitMargin	-0.055	Very weak negative relationship.

Correlation does not establish causation. The near-perfect quantity correlation should be interpreted carefully because it may reflect how the summary fields are aggregated or defined.

10. Conclusions and Recommended Follow-up

Focus	Recommended follow-up	Evidence to monitor
Supplier concentration	Review contingency suppliers and concentration by category.	Purchase contribution and cumulative share.
Inventory capital	Reconcile unsold inventory and investigate low-turnover vendors.	Verified on-hand value, age, and sell-through.
Brand promotion	Run controlled tests for low-sales/high-margin target brands.	Incremental gross profit and sell-through.
Bulk purchasing	Validate unit-price differences with landed cost and holding cost.	Net savings after carrying and freight costs.
Data quality	Document filtering, zero-denominator handling and outlier rules.	Reconciled metrics and repeatable pipeline.

Limitations

- All charts in this document are embedded from the supplied notebook outputs; no replacement or newly designed charts have been added.
- The main analyses use the notebook's filtered dataset, excluding rows with non-positive gross profit, profit margin, or total quantity.
- The report preserves notebook terminology and reported results; it does not rerun the database queries or independently verify the underlying transactions.
- Inventory value is an estimate produced by the notebook formula and should be reconciled before financial use.